

**Supplemental Material:**  
**Kitaev Honeycomb Model**

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This Supplemental Material is to present the details to prove personal understanding.

## SI. DERIVATIONS ABOUT THE COMMUTATOR AND EIGENVALUES

The plaquette operator has the equivalent expression:

$$\begin{aligned}
\hat{W}_p &= \hat{K}_{12} \hat{K}_{23} \hat{K}_{34} \hat{K}_{45} \hat{K}_{56} \hat{K}_{61} \\
&= \sigma_1^z \sigma_2^z \sigma_2^x \sigma_3^x \sigma_3^y \sigma_4^y \sigma_4^z \sigma_5^z \sigma_5^x \sigma_6^x \sigma_6^y \sigma_1^y \\
&= (\sigma_1^z \sigma_1^y) \sigma_2^z \sigma_2^x \sigma_3^x \sigma_3^y \sigma_4^y \sigma_4^z \sigma_5^z \sigma_5^x \sigma_6^x \sigma_6^y \\
&= (i\epsilon_{zyx} \sigma_1^x) (i\epsilon_{zxy} \sigma_2^y) (i\epsilon_{xyz} \sigma_3^z) (i\epsilon_{yzx} \sigma_4^x) (i\epsilon_{zxy} \sigma_5^y) (i\epsilon_{xyz} \sigma_6^z) \\
&= i^6 \epsilon_{zyx} \epsilon_{zxy} \epsilon_{xyz} \epsilon_{yzx} \epsilon_{zxy} \epsilon_{xyz} \sigma_1^x \sigma_2^y \sigma_3^z \sigma_4^x \sigma_5^y \sigma_6^z \\
&= (-1)(-1)(+1)(+1)(+1)(+1)(+1) \sigma_1^x \sigma_2^y \sigma_3^z \sigma_4^x \sigma_5^y \sigma_6^z \\
&= \sigma_1^x \sigma_2^y \sigma_3^z \sigma_4^x \sigma_5^y \sigma_6^z
\end{aligned} \tag{S1}$$

The square of the plaquette operator is

$$\hat{W}_p^2 = \prod_{j=1}^6 (\sigma_j^{\alpha_j})^2 = \hat{\mathbb{I}}$$

thus the eigenvalues of  $\hat{W}_p$  are  $\pm 1$ . For a bond on the hexagon, e.g. the  $z$ -bond  $\hat{K}_{12}$ , the bond label differs from the  $\hat{W}_p$  Pauli matrix at both shared sites:

$$\hat{K}_{12} \hat{W}_p = (\sigma_1^z \sigma_2^z) (\sigma_1^x \sigma_2^y \sigma_3^z \sigma_4^x \sigma_5^y \sigma_6^z) = (-1)^2 (\sigma_1^x \sigma_2^y \sigma_3^z \sigma_4^x \sigma_5^y \sigma_6^z) (\sigma_1^z \sigma_2^z) = \hat{W}_p \hat{K}_{12}.$$

For an outward bond, e.g. the  $x$ -bond  $\hat{K}_{11'}$  with  $1' \notin p$ , the bond label matches the Pauli matrix at the single shared site:

$$\hat{K}_{11'} \hat{W}_p = (\sigma_1^x \sigma_{1'}^x) (\sigma_1^x \sigma_2^y \sigma_3^z \sigma_4^x \sigma_5^y \sigma_6^z) = (+1) (\sigma_1^x \sigma_2^y \sigma_3^z \sigma_4^x \sigma_5^y \sigma_6^z) (\sigma_1^x \sigma_{1'}^x) = \hat{W}_p \hat{K}_{11'}.$$

A bond which does not touch  $p$  commutes trivially. Hence

$$[\hat{K}_{jk}, \hat{W}_p] = 0 \quad \implies \quad [\hat{H}, \hat{W}_p] = 0$$

So the Hilbert space splits into flux sectors labelled by the eigenvalues  $\{w_p\}$ . By four Majorana representation, the Hamiltonian is replaced by  $\tilde{H}\{\tilde{\sigma}_j\}$  and the bond operator becomes

$$\tilde{K}_{jk} = (i\hat{b}_j^\alpha \hat{\gamma}_j) (i\hat{b}_k^\alpha \hat{\gamma}_k) = -\hat{b}_j^\alpha \hat{\gamma}_j \hat{b}_k^\alpha \hat{\gamma}_k = \hat{b}_j^\alpha \hat{b}_k^\alpha \hat{\gamma}_j \hat{\gamma}_k = -i(\hat{b}_j^\alpha \hat{b}_k^\alpha) \hat{\gamma}_j \hat{\gamma}_k$$

thus the Hermitian Majorana bond operator is defined by:

$$\hat{u}_{jk} = i\hat{b}_j^\alpha \hat{b}_k^\alpha \quad \hat{u}_{jk}^\dagger = (i\hat{b}_j^\alpha \hat{b}_k^\alpha)^\dagger = -i\hat{b}_k^\alpha \hat{b}_j^\alpha = i\hat{b}_j^\alpha \hat{b}_k^\alpha = \hat{u}_{jk}.$$

The constraint operator  $\hat{D}_j = \hat{b}_j^x \hat{b}_j^y \hat{b}_j^z \hat{\gamma}_j$  commutes with the bond operator. A Majorana at site  $j$  commutes with its own copy inside  $\hat{D}_j$  and anticommutes with the other three factors. A Majorana at  $k \neq j$  anticommutes with all four:

$$\begin{aligned} \hat{b}_j^\alpha \hat{D}_j &= (-1)^3 \hat{D}_j \hat{b}_j^\alpha, & \hat{\gamma}_j \hat{D}_j &= (-1)^3 \hat{D}_j \hat{\gamma}_j, \\ \hat{b}_k^\alpha \hat{D}_j &= (-1)^4 \hat{D}_j \hat{b}_k^\alpha, & \hat{\gamma}_k \hat{D}_j &= (-1)^4 \hat{D}_j \hat{\gamma}_k. \end{aligned}$$

Hence

$$\begin{aligned} \hat{D}_j \tilde{K}_{jk} &= \hat{D}_j \hat{b}_j^\alpha \hat{b}_k^\alpha \hat{\gamma}_j \hat{\gamma}_k \\ &= -\hat{b}_j^\alpha \hat{D}_j \hat{b}_k^\alpha \hat{\gamma}_j \hat{\gamma}_k \\ &= -\hat{b}_j^\alpha \hat{b}_k^\alpha \hat{D}_j \hat{\gamma}_j \hat{\gamma}_k \\ &= +\hat{b}_j^\alpha \hat{b}_k^\alpha \hat{\gamma}_j \hat{D}_j \hat{\gamma}_k \\ &= \hat{b}_j^\alpha \hat{b}_k^\alpha \hat{\gamma}_j \hat{\gamma}_k \hat{D}_j = \tilde{K}_{jk} \hat{D}_j, \end{aligned}$$

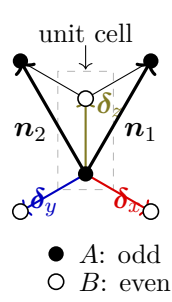
and a bond not containing  $j$  commutes with  $\hat{D}_j$  trivially, so

$$[\hat{D}_j, \tilde{K}_{jk}] = 0 \quad \implies \quad [\hat{D}_j, \tilde{H}] = 0, \tag{S2}$$

i.e.  $\tilde{H}$  preserves the physical subspace  $\hat{D}_j |\xi\rangle = |\xi\rangle$ .

### SII. LATTICE CONVENTIONS

The system is translation invariant under any Bravais vector  $\mathbf{R} = n_1 \mathbf{n}_1 + n_2 \mathbf{n}_2$  with  $n_1, n_2 \in \mathbb{Z}$ . We place the odd ( $A$ ) sublattice at the cell origin  $\mathbf{R}$  and the even ( $B$ ) sublattice at  $\mathbf{R} + \boldsymbol{\delta}_z$ . The nearest-neighbour bond vectors  $\boldsymbol{\delta}_\alpha$  point from  $A$  to  $B$ , and the primitive vectors are  $\mathbf{n}_1 = \boldsymbol{\delta}_z - \boldsymbol{\delta}_y$ ,  $\mathbf{n}_2 = \boldsymbol{\delta}_z - \boldsymbol{\delta}_x$  (Fig. S1). Note  $\boldsymbol{\delta}_x + \boldsymbol{\delta}_y + \boldsymbol{\delta}_z = 0$ .



$$\begin{aligned}\delta_x &= \left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right), \\ \delta_y &= \left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right), \\ \delta_z &= (0, 1), \\ n_1 &= \left(\frac{\sqrt{3}}{2}, \frac{3}{2}\right), \\ n_2 &= \left(-\frac{\sqrt{3}}{2}, \frac{3}{2}\right).\end{aligned}$$

FIG. S1. Bond vectors  $\delta_\alpha$  ( $A \rightarrow B$ , coloured by bond type) and primitive Bravais vectors  $\mathbf{n}_{1,2}$  of the honeycomb lattice, with the two-site unit cell (dashed).

### III. FOURIER TRANSFORM, BLOCH HAMILTONIAN, AND THE DIRAC CONES

The real-space Hamiltonian in the  $u_{jk} = 1$  gauge is

$$\tilde{H}_u = i \sum_{\mathbf{R}, \alpha} J_\alpha \hat{\gamma}_{\mathbf{R}}^A \hat{\gamma}_{\mathbf{R}+\delta_\alpha}^B. \quad (\text{S3})$$

The prefactor  $i$  comes from the quadratic form  $\frac{i}{4} \sum_{j,k} A_{jk} \hat{\gamma}_j \hat{\gamma}_k$  with  $A_{jk} = 2J_\alpha u_{jk}$ ,  $u_{jk} = +1$ , after removing the double counting of each bond. Here I use the *periodic gauge* for the Fourier transform, so each Majorana carries the phase of its physical position:

$$\hat{\gamma}_{\mathbf{R}}^A = \frac{1}{\sqrt{N}} \sum_{\mathbf{q}} e^{i\mathbf{q} \cdot \mathbf{R}} \tilde{\gamma}_{\mathbf{q}}^A, \quad \hat{\gamma}_{\mathbf{R}+\delta_\alpha}^B = \frac{1}{\sqrt{N}} \sum_{\mathbf{q}} e^{i\mathbf{q} \cdot (\mathbf{R}+\delta_\alpha)} \tilde{\gamma}_{\mathbf{q}}^B, \quad (\tilde{\gamma}_{\mathbf{q}}^{A,B})^\dagger = \tilde{\gamma}_{-\mathbf{q}}^{A,B}. \quad (\text{S4})$$

Substitute and sum over  $\mathbf{R}$ ,

$$\begin{aligned}\tilde{H}_u &= \frac{i}{N} \sum_{\mathbf{R}, \alpha} J_\alpha \sum_{\mathbf{q}, \mathbf{q}'} e^{i\mathbf{q} \cdot \mathbf{R}} e^{i\mathbf{q}' \cdot (\mathbf{R}+\delta_\alpha)} \tilde{\gamma}_{\mathbf{q}}^A \tilde{\gamma}_{\mathbf{q}'}^B \\ &= \frac{i}{N} \sum_{\alpha} J_\alpha \sum_{\mathbf{q}, \mathbf{q}'} e^{i\mathbf{q}' \cdot \delta_\alpha} \tilde{\gamma}_{\mathbf{q}}^A \tilde{\gamma}_{\mathbf{q}'}^B \underbrace{\sum_{\mathbf{R}} e^{i(\mathbf{q}+\mathbf{q}') \cdot \mathbf{R}}}_{N \delta_{\mathbf{q}+\mathbf{q}', 0}} = i \sum_{\alpha} J_\alpha \sum_{\mathbf{q}} e^{-i\mathbf{q} \cdot \delta_\alpha} \tilde{\gamma}_{\mathbf{q}}^A \tilde{\gamma}_{-\mathbf{q}}^B.\end{aligned} \quad (\text{S5})$$

We split the  $\mathbf{q}$ -sum over  $\mathbf{q}$  and  $-\mathbf{q}$  and use  $\tilde{\gamma}_{-\mathbf{q}}^A \tilde{\gamma}_{\mathbf{q}}^B = -\tilde{\gamma}_{\mathbf{q}}^B \tilde{\gamma}_{-\mathbf{q}}^A$ . This gives a Hermitian form

$$\begin{aligned}\tilde{H}_u &= \frac{i}{2} \sum_{\alpha} J_\alpha \sum_{\mathbf{q}} \left( e^{-i\mathbf{q} \cdot \delta_\alpha} \tilde{\gamma}_{\mathbf{q}}^A \tilde{\gamma}_{-\mathbf{q}}^B + e^{i\mathbf{q} \cdot \delta_\alpha} \tilde{\gamma}_{-\mathbf{q}}^A \tilde{\gamma}_{\mathbf{q}}^B \right) \\ &= \frac{i}{2} \sum_{\alpha} J_\alpha \sum_{\mathbf{q}} \left( e^{-i\mathbf{q} \cdot \delta_\alpha} \tilde{\gamma}_{\mathbf{q}}^A \tilde{\gamma}_{-\mathbf{q}}^B - e^{i\mathbf{q} \cdot \delta_\alpha} \tilde{\gamma}_{\mathbf{q}}^B \tilde{\gamma}_{-\mathbf{q}}^A \right).\end{aligned} \quad (\text{S6})$$

Choosing the sublattice spinor

$$\chi_{\mathbf{q}} = \begin{pmatrix} \tilde{\gamma}_{\mathbf{q}}^A \\ \tilde{\gamma}_{\mathbf{q}}^B \end{pmatrix}, \quad \chi_{\mathbf{q}}^\dagger = \begin{pmatrix} \tilde{\gamma}_{-\mathbf{q}}^A & \tilde{\gamma}_{-\mathbf{q}}^B \end{pmatrix}, \quad (\text{S7})$$

the Hamiltonian takes the Bloch form of the main text,

$$\tilde{H}_u = \frac{1}{2} \sum_{\mathbf{q}} \chi_{\mathbf{q}}^\dagger h(\mathbf{q}) \chi_{\mathbf{q}}, \quad h(\mathbf{q}) = \begin{pmatrix} 0 & \text{i}f(\mathbf{q}) \\ -\text{i}f(\mathbf{q})^* & 0 \end{pmatrix}, \quad (\text{S8})$$

with

$$f(\mathbf{q}) = \sum_{\alpha} J_{\alpha} e^{\text{i}\mathbf{q} \cdot \boldsymbol{\delta}_{\alpha}} = J_x e^{\text{i}(\sqrt{3}q_x - q_y)/2} + J_y e^{-\text{i}(\sqrt{3}q_x + q_y)/2} + J_z e^{\text{i}q_y}. \quad (\text{S9})$$

The bulk dispersion  $\epsilon_{\pm}(\mathbf{q}) = \pm |f(\mathbf{q})|$  is explicitly

$$|f(\mathbf{q})|^2 = J_x^2 + J_y^2 + J_z^2 + 2J_x J_y \cos(\sqrt{3}q_x) + 2J_x J_z \cos\left(\frac{\sqrt{3}q_x - 3q_y}{2}\right) + 2J_y J_z \cos\left(\frac{\sqrt{3}q_x + 3q_y}{2}\right). \quad (\text{S10})$$

Expanding  $h(\mathbf{q}) = \mathbf{d}(\mathbf{q}) \cdot \boldsymbol{\sigma}$ , the components:

$$\begin{aligned} d_x(\mathbf{q}) &= -\text{Im} f(\mathbf{q}) = -\sum_{\alpha} J_{\alpha} \sin(\mathbf{q} \cdot \boldsymbol{\delta}_{\alpha}), \\ d_y(\mathbf{q}) &= -\text{Re} f(\mathbf{q}) = -\sum_{\alpha} J_{\alpha} \cos(\mathbf{q} \cdot \boldsymbol{\delta}_{\alpha}), \quad d_z(\mathbf{q}) = 0. \end{aligned} \quad (\text{S11})$$

The vanishing of  $d_z$  is the chiral symmetry  $\sigma_z h(\mathbf{q}) \sigma_z = -h(\mathbf{q})$ :  $\mathbf{d}$  stays in the  $xy$ -plane over the whole Brillouin zone. Equivalently  $d_x + \text{i}d_y = -\text{i}f^*(\mathbf{q})$ .

At the symmetric point  $J_x = J_y = J_z \equiv J$ ,  $f = 0$  gives

$$e^{\text{i}\phi_x} + e^{\text{i}\phi_y} + e^{\text{i}\phi_z} = 0, \quad \phi_{\alpha} \equiv \mathbf{q} \cdot \boldsymbol{\delta}_{\alpha} : \quad (\text{S12})$$

three unit vectors summing to zero form an equilateral triangle, so the phases differ by  $\pm 2\pi/3$ . We choose

$$\phi_x = -\frac{2\pi}{3}, \quad \phi_y = 0, \quad \phi_z = \frac{2\pi}{3}. \quad (\text{S13})$$

Other assignments give the same points up to reciprocal-lattice vectors. The component equations  $\mathbf{q} \cdot \boldsymbol{\delta}_z = q_y = \frac{2\pi}{3}$  and  $\mathbf{q} \cdot \boldsymbol{\delta}_y = -\frac{\sqrt{3}}{2}q_x - \frac{1}{2}q_y = 0$  give

$$\mathbf{q}^* \equiv \pm \mathbf{K}, \quad \mathbf{K} = \left( -\frac{2\pi}{3\sqrt{3}}, \frac{2\pi}{3} \right), \quad (\text{S14})$$

the two inequivalent corners of the Brillouin zone.

Expanding  $f(\mathbf{K} + \delta\mathbf{q}) \simeq f(\mathbf{K}) + \delta\mathbf{q} \cdot \nabla f|_{\mathbf{K}}$  with

$$\nabla f(\mathbf{q}) = iJ \sum_{\alpha} \delta_{\alpha} e^{i\mathbf{q} \cdot \delta_{\alpha}}, \quad (\text{S15})$$

we evaluate at  $\mathbf{K}$  using the three phases above:

$$\begin{aligned} \nabla f|_{\mathbf{K}} &= iJ \left( \delta_x e^{-i2\pi/3} + \delta_y + \delta_z e^{i2\pi/3} \right) \\ &= iJ \left[ \left( \frac{\sqrt{3}}{2}, -\frac{1}{2} \right) \left( -\frac{1}{2} - i\frac{\sqrt{3}}{2} \right) + \left( -\frac{\sqrt{3}}{2}, -\frac{1}{2} \right) + (0, 1) \left( -\frac{1}{2} + i\frac{\sqrt{3}}{2} \right) \right] \\ &= \frac{3J}{4} (1 - \sqrt{3}i, -\sqrt{3} - i). \end{aligned} \quad (\text{S16})$$

Hence, with  $f(\mathbf{K}) = 0$ ,

$$\begin{aligned} f(\mathbf{K} + \delta\mathbf{q}) &\simeq \frac{3J}{4} \left[ (1 - \sqrt{3}i) \delta q_x + (-\sqrt{3} - i) \delta q_y \right] \\ &= \frac{3J}{2} \left( \frac{1}{2} - i\frac{\sqrt{3}}{2} \right) (\delta q_x - i \delta q_y) = \frac{3J}{2} e^{-i\pi/3} (\delta q_x - i \delta q_y). \end{aligned} \quad (\text{S17})$$

The same expansion at  $-\mathbf{K}$  gives  $\delta q_x + i \delta q_y$ , since  $f(-\mathbf{q}) = f(\mathbf{q})^*$ . The dispersion is an isotropic Dirac cone

$$\epsilon(\mathbf{K} + \delta\mathbf{q}) \simeq \frac{3J}{2} |\delta\mathbf{q}|, \quad v_F = \frac{3J}{2}, \quad (\text{S18})$$

as in graphene (Fig. S2).

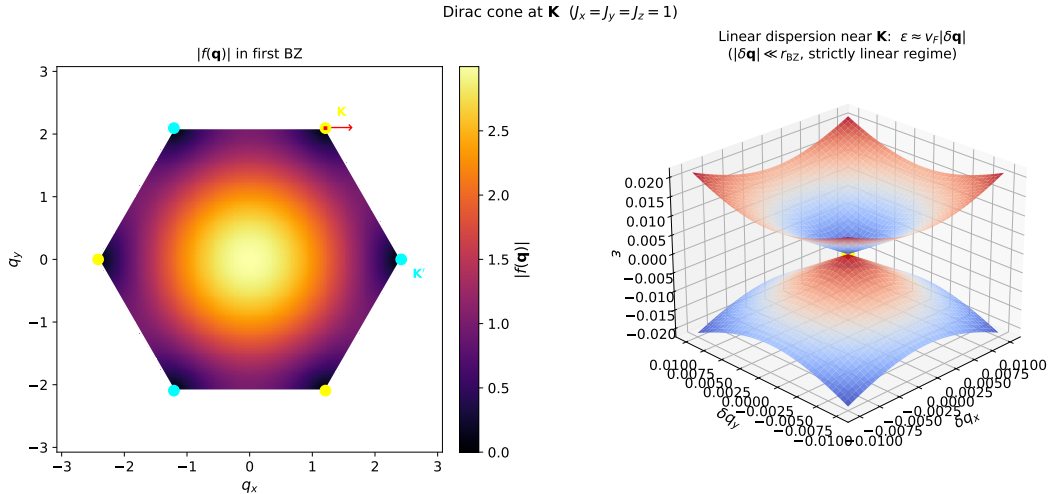


FIG. S2. Dirac cone at the Brillouin-zone corner  $\mathbf{K}$  for  $J_x = J_y = J_z$ . Left:  $|f(\mathbf{q})|$  over the first BZ, vanishing at the six corners. Right: the linear dispersion  $\epsilon(\mathbf{K} + \delta\mathbf{q}) \approx v_F |\delta\mathbf{q}|$  near  $\mathbf{K}$ , as in graphene.

#### SIV. TIME REVERSAL: FROM THE SPINS TO THE BLOCH CONSTRAINT

In this section I want to show how the spin-level time reversal becomes the Bloch-level constraint  $\hat{T} = \sigma_z \mathcal{K}$ . Time reversal acts on spins as  $\hat{T} \sigma_j^\alpha \hat{T}^{-1} = -\sigma_j^\alpha$ . Each bond term  $\sigma_j^\alpha \sigma_k^\alpha$  is quadratic in spins and picks up two minus signs, so it is invariant. The Kitaev Hamiltonian is time-reversal symmetric [1].

In the four-Majorana representation  $\tilde{\sigma}^\alpha = i \hat{b}^\alpha \hat{\gamma}$ ,

$$\hat{T} \tilde{\sigma}^\alpha \hat{T}^{-1} = -\tilde{\sigma}^\alpha \implies \hat{T} (\hat{b}^\alpha \hat{\gamma}) \hat{T}^{-1} = +\hat{b}^\alpha \hat{\gamma}, \quad (\text{S19})$$

so on each site all four Majoranas are simultaneously  $\hat{T}$ -odd or  $\hat{T}$ -even. Both choices leave  $\hat{D}_j = \hat{b}_j^x \hat{b}_j^y \hat{b}_j^z \hat{\gamma}_j$  and the physical subspace invariant. But the vortex-free Hamiltonian Eq. (S3) fixes the choice on each sublattice. Writing  $\hat{T} \hat{\gamma}^A \hat{T}^{-1} = s_A \hat{\gamma}^A$ ,  $\hat{T} \hat{\gamma}^B \hat{T}^{-1} = s_B \hat{\gamma}^B$ ,

$$\hat{T} (i J_\alpha \hat{\gamma}^A \hat{\gamma}^B) \hat{T}^{-1} = -s_A s_B i J_\alpha \hat{\gamma}^A \hat{\gamma}^B = i J_\alpha \hat{\gamma}^A \hat{\gamma}^B \implies s_A s_B = -1. \quad (\text{S20})$$

We choose

$$\hat{T} \hat{\gamma}_{\mathbf{R}}^A \hat{T}^{-1} = -\hat{\gamma}_{\mathbf{R}}^A, \quad \hat{T} \hat{\gamma}_{\mathbf{R}}^B \hat{T}^{-1} = +\hat{\gamma}_{\mathbf{R}}^B. \quad (\text{S21})$$

Using the inverse Fourier transformation,  $\tilde{\gamma}_{\mathbf{q}}^A = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} e^{-i\mathbf{q} \cdot \mathbf{R}} \hat{\gamma}_{\mathbf{R}}^A$ , which follows

$$\hat{T} \tilde{\gamma}_{\mathbf{q}}^A \hat{T}^{-1} = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} e^{+i\mathbf{q} \cdot \mathbf{R}} (\hat{T} \hat{\gamma}_{\mathbf{R}}^A \hat{T}^{-1}) = -\frac{1}{\sqrt{N}} \sum_{\mathbf{R}} e^{i\mathbf{q} \cdot \mathbf{R}} \hat{\gamma}_{\mathbf{R}}^A = -\tilde{\gamma}_{-\mathbf{q}}^A, \quad \hat{T} \tilde{\gamma}_{\mathbf{q}}^B \hat{T}^{-1} = +\tilde{\gamma}_{-\mathbf{q}}^B. \quad (\text{S22})$$

On the spinor Eq. (S7) this is

$$\hat{T} \chi_{\mathbf{q}} \hat{T}^{-1} = \sigma_z \chi_{-\mathbf{q}}, \quad \hat{T} \chi_{\mathbf{q}}^\dagger \hat{T}^{-1} = \chi_{-\mathbf{q}}^\dagger \sigma_z : \quad (\text{S23})$$

at the Bloch level,  $\hat{T} = \sigma_z \mathcal{K}$  with  $\mathcal{K}$  complex conjugation. This could also be directly realized by the isomorphism of physical subspace and spin space.

Apply the  $\hat{T}$  on the Bloch Hamiltonian Eq. (S8)

$$\hat{T} \tilde{H}_u \hat{T}^{-1} = \frac{1}{2} \sum_{\mathbf{q}} \chi_{-\mathbf{q}}^\dagger \sigma_z h^*(\mathbf{q}) \sigma_z \chi_{-\mathbf{q}} = \frac{1}{2} \sum_{\mathbf{q}} \chi_{\mathbf{q}}^\dagger \sigma_z h^*(-\mathbf{q}) \sigma_z \chi_{\mathbf{q}}, \quad (\text{S24})$$

where the last step relabels  $\mathbf{q} \rightarrow -\mathbf{q}$ . Comparing with  $\tilde{H}_u = \frac{1}{2} \sum_{\mathbf{q}} \chi_{\mathbf{q}}^\dagger h(\mathbf{q}) \chi_{\mathbf{q}}$ , we get the time-reversal constraint

$$\sigma_z h^*(-\mathbf{q}) \sigma_z = h(\mathbf{q}). \quad (\text{S25})$$

Writing  $h = d_x\sigma_x + d_y\sigma_y + d_z\sigma_z$  and using  $\sigma_{x,z}^* = \sigma_{x,z}$ ,  $\sigma_y^* = -\sigma_y$ ,  $\sigma_z\sigma_{x,y}\sigma_z = -\sigma_{x,y}$ , the left-hand side equals  $-d_x(-\mathbf{q})\sigma_x + d_y(-\mathbf{q})\sigma_y + d_z(-\mathbf{q})\sigma_z$ . Matching components,

$$d_x(\mathbf{q}) = -d_x(-\mathbf{q}), \quad d_y(\mathbf{q}) = +d_y(-\mathbf{q}), \quad d_z(\mathbf{q}) = +d_z(-\mathbf{q}). \quad (\text{S26})$$

The Bloch Hamiltonian obeys these automatically:  $f(-\mathbf{q}) = f(\mathbf{q})^*$  makes  $d_x = -\text{Im } f$  odd and  $d_y = -\text{Re } f$  even. A time-reversal-symmetric perturbation can only generate an even mass  $d_z(\mathbf{q})$ . An even mass gives the two Dirac points equal masses, then the Chern number vanishes (Section SVIII). A topological gap needs an odd  $d_z$ , the time-reversal breaking mass, so time reversal must be broken.

## SV. PERTURBATION THEORY: ORDER BY ORDER

The Zeeman perturbation is  $V = -\sum_j \mathbf{h} \cdot \boldsymbol{\sigma}_j$ . For the effective Hamiltonian in the vortex-free sector we use the Schrieffer–Wolff expansion

$$H_{\text{eff}} = \sum_{n \geq 1} H_{\text{eff}}^{(n)}, \quad H_{\text{eff}}^{(n)} = \Pi_0 V (G'_0 V)^{n-1} \Pi_0, \quad \Pi_0 = \prod_p \frac{1 + \hat{W}_p}{2} \prod_j \frac{1 + \hat{D}_j}{2}, \quad (\text{S27})$$

where  $G'_0$  is the unperturbed Green's function with the vortex-free sector excluded. The intermediate states carry a vortex pair. The vortex pair costs an energy of order  $|J|$ . We approximate all intermediate energies by this scale:

$$G'_0(E_0) \simeq -\frac{1 - \Pi_0}{|J|}. \quad (\text{S28})$$

$V$  carries one factor of  $\mathbf{h} \cdot \boldsymbol{\sigma}$ , so  $\hat{T}V\hat{T}^{-1} = -V$ , while  $\Pi_0$  and  $G'_0$  are  $\hat{T}$ -invariant. The  $n$ -th order term then transforms as  $\hat{T}H_{\text{eff}}^{(n)}\hat{T}^{-1} = (-1)^n H_{\text{eff}}^{(n)}$ . Only odd orders can generate the time-reversal breaking mass required by Eq. (S26).

*First order.*

$$H_{\text{eff}}^{(1)} = \Pi_0 V \Pi_0 = 0. \quad (\text{S29})$$

A single Zeeman operator  $\sigma_j^\alpha = i\hat{b}_j^\alpha \hat{\gamma}_j$  carries one gauge Majorana  $\hat{b}_j^\alpha$ . It anticommutes with the link operator  $\hat{u}_{jk} = i\hat{b}_j^\alpha \hat{b}_k^\alpha$  on the  $\alpha$ -bond at site  $j$ :  $\{\hat{b}_j^\alpha, \hat{u}_{jk}\} = 0$ . So it flips  $u_{jk} \rightarrow -u_{jk}$ . The bond is shared by two plaquettes, so both adjacent  $w_p$  flip from  $+1$  to  $-1$  (Fig. S3, second panel). The state leaves the vortex-free sector, and the outer  $\Pi_0$  projects it to zero.

Second order.

$$H_{\text{eff}}^{(2)} = \Pi_0 V G'_0 V \Pi_0 \neq 0. \quad (\text{S30})$$

This term does not vanish: the first Zeeman operator creates a vortex pair and the second one can annihilate the same pair. Both  $w_p$  return to +1 and the state passes the projectors. However, with  $n = 2$  factors of  $\mathbf{h}$  the term is even under time reversal. By Eq. (S26) it can only produce an even  $d_z$  or renormalize the existing couplings; it can not make the gap topological, so we drop it.

Third order.

$$H_{\text{eff}}^{(3)} = \Pi_0 V G'_0 V G'_0 V \Pi_0 = \frac{1}{|J|^2} \Pi_0 V (1 - \Pi_0) V (1 - \Pi_0) V \Pi_0, \quad (\text{S31})$$

using Eq. (S28). This is the leading time-reversal breaking term. Only the processes which return all  $w_p$  to +1 remain in the vortex-free sector. As shown in Fig. S3, the only possible configuration is three spin operators on the three different bond directions around one site. Each intermediate state carries exactly one vortex pair. With three factors of  $\mathbf{h}$  and two energy denominators,

$$H_{\text{eff}}^{(3)} = -\kappa \sum_{\langle jkl \rangle} \sigma_j^x \sigma_k^y \sigma_l^z, \quad \kappa \sim \frac{h^x h^y h^z}{|J|^2}. \quad (\text{S32})$$

In each triple the central spin  $\sigma_l^z$  sits on an even ( $B$ ) site  $l$ . The spins  $\sigma_j^x, \sigma_k^y$  sit on the two odd ( $A$ ) sites joined to  $l$  by its  $x$ - and  $y$ -bonds (Fig. S4).

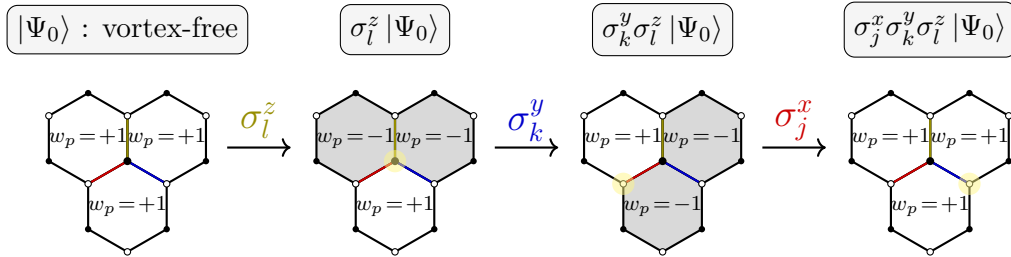


FIG. S3. The flux-restoring third-order process. Start from the vortex-free state  $|\Psi_0\rangle$  (all  $w_p = +1$ , white). Each Zeeman operator  $\sigma_l^\alpha$  at the highlighted site flips the two  $w_p$  next to its  $\alpha$ -bond (grey,  $w_p = -1$ ). Only the full product  $\sigma_j^x \sigma_k^y \sigma_l^z$  returns all  $w_p$  to +1; the intermediate one- and two-operator states lie outside the vortex-free sector and are projected out by  $\Pi_0$ . Bond colours:  $x$  red,  $y$  blue,  $z$  olive.

## SVI. REDUCTION OF THE THREE-SPIN TERM TO A MAJORANA BILINEAR

Insert the Majorana representation  $\sigma_j^\alpha = i \hat{b}_j^\alpha \hat{\gamma}_j$  into one triple of Eq. (S32):

$$\sigma_j^x \sigma_k^y \sigma_l^z = (i \hat{b}_j^x \hat{\gamma}_j)(i \hat{b}_k^y \hat{\gamma}_k)(i \hat{b}_l^z \hat{\gamma}_l) = -i \hat{b}_j^x \hat{\gamma}_j \hat{b}_k^y \hat{\gamma}_k \hat{b}_l^z \hat{\gamma}_l. \quad (\text{S33})$$

On the central site  $l$  we reduce the pair  $\hat{b}_l^z \hat{\gamma}_l$  by the physical-subspace condition  $\hat{D}_l = \hat{b}_l^x \hat{b}_l^y \hat{b}_l^z \hat{\gamma}_l = 1$ . Moving  $\hat{\gamma}_l$  through the three  $\hat{b}_l$ 's gives  $(-1)^3$ , and  $\hat{\gamma}_l^2 = 1$ , so

$$\hat{b}_l^z \hat{\gamma}_l = \hat{b}_l^z \hat{\gamma}_l \hat{D}_l = \hat{b}_l^z \hat{\gamma}_l \hat{b}_l^x \hat{b}_l^y \hat{b}_l^z \hat{\gamma}_l = -\hat{b}_l^z \hat{b}_l^x \hat{b}_l^y \hat{b}_l^z = -\hat{b}_l^x \hat{b}_l^y, \quad (\text{S34})$$

where in the last step the first  $\hat{b}_l^z$  passes through  $\hat{b}_l^x \hat{b}_l^y$  (two anticommutations, sign +1) and  $(\hat{b}_l^z)^2 = 1$ . Hence

$$\sigma_j^x \sigma_k^y \sigma_l^z = +i \hat{b}_j^x \hat{\gamma}_j \hat{b}_k^y \hat{\gamma}_k \hat{b}_l^x \hat{b}_l^y = +i (\hat{b}_j^x \hat{b}_l^x) (\hat{b}_k^y \hat{b}_l^y) (\hat{\gamma}_j \hat{\gamma}_k). \quad (\text{S35})$$

The reordering is an even permutation of six Majorana operators, so the sign is +1. I track this sign carefully, since it fixes the sign of the final hopping term. With the link operators  $\hat{u}_{jl} = i \hat{b}_j^x \hat{b}_l^x$  and  $\hat{u}_{kl} = i \hat{b}_k^y \hat{b}_l^y$  (so  $\hat{b}_j^x \hat{b}_l^x = -i \hat{u}_{jl}$ ,  $\hat{b}_k^y \hat{b}_l^y = -i \hat{u}_{kl}$ ),

$$\sigma_j^x \sigma_k^y \sigma_l^z = i (-i \hat{u}_{jl}) (-i \hat{u}_{kl}) \hat{\gamma}_j \hat{\gamma}_k = -i \hat{u}_{jl} \hat{u}_{kl} \hat{\gamma}_j \hat{\gamma}_k \xrightarrow{u_{jl}=u_{kl}=1} -i \hat{\gamma}_j \hat{\gamma}_k. \quad (\text{S36})$$

The three-spin operator is now a next-nearest-neighbour Majorana hopping between the two same-sublattice ( $A$ ) sites  $j, k$ , displaced by  $\mathbf{n}_1 - \mathbf{n}_2$  (Fig. S4). The triples from the other two bond pairs around a site give the displacements  $\mathbf{n}_1$  and  $\mathbf{n}_2$ . This imaginary second-neighbour hopping has the same structure as the Haldane model [2], here for Majorana fermions.

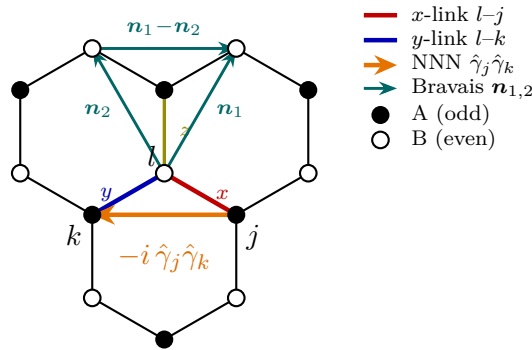


FIG. S4. Next-nearest-neighbour Majorana hopping from the third-order perturbation. The three-spin operator  $\sigma_j^x \sigma_k^y \sigma_l^z$  acts on a triple sharing the central even ( $B$ ) site  $l$ , joined by its  $x$ - and  $y$ -links to the two odd ( $A$ ) sites  $j, k$ . In the fixed gauge  $u_{jk} = 1$  it becomes a next-nearest-neighbour hopping  $-i \hat{\gamma}_j \hat{\gamma}_k$  (orange) between the two same-sublattice sites  $j, k$ , displaced by  $\mathbf{n}_1 - \mathbf{n}_2$ .

## SVII. THE MASS TERM $d_z(\mathbf{q})$ AND THE GAP

Write  $\hat{\gamma}_j \rightarrow \hat{\gamma}_{\mathbf{R}}^A$  and  $\hat{\gamma}_k \rightarrow \hat{\gamma}_{\mathbf{R}+\mathbf{n}}^A$  with  $\mathbf{n} \in \{\mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_1 - \mathbf{n}_2\}$ . An  $A$ -centred process carries the opposite sign on the  $B$  sublattice, since the directed bond reverses:  $\hat{u}_{l'l'} = -\hat{u}_{j'j'}$ . The third-order contribution then takes the same  $\frac{i}{4} \sum A_{jk} \hat{\gamma}_j \hat{\gamma}_k$  form as the nearest-neighbour Hamiltonian:

$$H_{\text{eff}}^{(3)} = \frac{i}{4} \kappa \sum_{\mathbf{R}} \sum_{\mathbf{n}} [\hat{\gamma}_{\mathbf{R}}^A \hat{\gamma}_{\mathbf{R}+\mathbf{n}}^A - \hat{\gamma}_{\mathbf{R}}^B \hat{\gamma}_{\mathbf{R}+\mathbf{n}}^B], \quad \mathbf{n} \in \{\mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_1 - \mathbf{n}_2\}. \quad (\text{S37})$$

Both operators of a same-sublattice term share the offset  $\delta_z$  (or none). The shared offset cancels under momentum conservation, so we transform with the cell label only,  $\hat{\gamma}_{\mathbf{R}}^A = \frac{1}{\sqrt{N}} \sum_{\mathbf{q}} e^{i\mathbf{q} \cdot \mathbf{R}} \tilde{\gamma}_{\mathbf{q}}^A$ . Substituting,

$$\begin{aligned} H^{AA} &= \kappa \frac{i}{4} \sum_{\mathbf{n}} \sum_{\mathbf{R}} \hat{\gamma}_{\mathbf{R}}^A \hat{\gamma}_{\mathbf{R}+\mathbf{n}}^A = \kappa \frac{i}{4} \sum_{\mathbf{n}} \frac{1}{N} \sum_{\mathbf{q}, \mathbf{q}'} e^{i\mathbf{q}' \cdot \mathbf{n}} \tilde{\gamma}_{\mathbf{q}}^A \tilde{\gamma}_{\mathbf{q}'}^A \underbrace{\sum_{\mathbf{R}} e^{i(\mathbf{q}+\mathbf{q}') \cdot \mathbf{R}}}_{N \delta_{\mathbf{q}+\mathbf{q}', 0}} \\ &= \kappa \frac{i}{4} \sum_{\mathbf{n}} \sum_{\mathbf{q}} e^{-i\mathbf{q} \cdot \mathbf{n}} \tilde{\gamma}_{\mathbf{q}}^A \tilde{\gamma}_{-\mathbf{q}}^A. \end{aligned} \quad (\text{S38})$$

The two operators now live on the same sublattice, so  $\tilde{\gamma}_{-\mathbf{q}}^A \tilde{\gamma}_{\mathbf{q}}^A = -\tilde{\gamma}_{\mathbf{q}}^A \tilde{\gamma}_{-\mathbf{q}}^A$ . The bilinear is odd under  $\mathbf{q} \rightarrow -\mathbf{q}$ . We symmetrize the  $\mathbf{q}$ -sum by relabelling  $\mathbf{q} \rightarrow -\mathbf{q}$  in one half:

$$\begin{aligned} \sum_{\mathbf{q}} e^{-i\mathbf{q} \cdot \mathbf{n}} \tilde{\gamma}_{\mathbf{q}}^A \tilde{\gamma}_{-\mathbf{q}}^A &= \frac{1}{2} \sum_{\mathbf{q}} \left( e^{-i\mathbf{q} \cdot \mathbf{n}} \tilde{\gamma}_{\mathbf{q}}^A \tilde{\gamma}_{-\mathbf{q}}^A + e^{+i\mathbf{q} \cdot \mathbf{n}} \tilde{\gamma}_{-\mathbf{q}}^A \tilde{\gamma}_{\mathbf{q}}^A \right) \\ &= \frac{1}{2} \sum_{\mathbf{q}} (e^{-i\mathbf{q} \cdot \mathbf{n}} - e^{+i\mathbf{q} \cdot \mathbf{n}}) \tilde{\gamma}_{\mathbf{q}}^A \tilde{\gamma}_{-\mathbf{q}}^A = -i \sum_{\mathbf{q}} \sin(\mathbf{q} \cdot \mathbf{n}) \tilde{\gamma}_{\mathbf{q}}^A \tilde{\gamma}_{-\mathbf{q}}^A. \end{aligned} \quad (\text{S39})$$

In the nearest-neighbour  $A$ - $B$  block the sublattices are distinct, so the full exponential  $f(\mathbf{q})$  survives. The relevant dot products are

$$\mathbf{q} \cdot \mathbf{n}_1 = \frac{\sqrt{3}}{2} q_x + \frac{3}{2} q_y, \quad \mathbf{q} \cdot \mathbf{n}_2 = -\frac{\sqrt{3}}{2} q_x + \frac{3}{2} q_y, \quad \mathbf{q} \cdot (\mathbf{n}_1 - \mathbf{n}_2) = \sqrt{3} q_x. \quad (\text{S40})$$

Thus  $H^{AA} = \frac{\kappa}{4} \sum_{\mathbf{q}} [\sum_{\mathbf{n}} \sin(\mathbf{q} \cdot \mathbf{n})] \tilde{\gamma}_{\mathbf{q}}^A \tilde{\gamma}_{-\mathbf{q}}^A$ . Plugging into the Bloch form  $\frac{1}{2} \sum_{\mathbf{q}} \chi_{\mathbf{q}}^\dagger h(\mathbf{q}) \chi_{\mathbf{q}}$ , the upper-left ( $A$ - $A$ ) diagonal entry gives  $d_z(\mathbf{q})$  up to a positive  $O(1)$  factor. The coupling  $\kappa \sim h^x h^y h^z / |J|^2$  is only fixed up to such a factor. We absorb this factor and the overall sign into  $\kappa$ . The mass term is

$$d_z(\mathbf{q}) = 6\kappa \left[ \sin(\mathbf{q} \cdot \mathbf{n}_1) + \sin(\mathbf{q} \cdot \mathbf{n}_2) + \sin(\mathbf{q} \cdot (\mathbf{n}_1 - \mathbf{n}_2)) \right]. \quad (\text{S41})$$

The  $B$ -sublattice block carries the opposite sign in Eq. (S37) and gives the lower-right entry  $-d_z(\mathbf{q})$ . Together the two blocks give  $d_z(\mathbf{q}) \sigma_z$ , and the Bloch Hamiltonian becomes

$$h(\mathbf{q}) = \begin{pmatrix} d_z(\mathbf{q}) & if(\mathbf{q}) \\ -if^*(\mathbf{q}) & -d_z(\mathbf{q}) \end{pmatrix} = \mathbf{d}(\mathbf{q}) \cdot \boldsymbol{\sigma}, \quad \varepsilon_{\pm}(\mathbf{q}) = \pm \sqrt{|f(\mathbf{q})|^2 + d_z(\mathbf{q})^2}, \quad (\text{S42})$$

since  $d_x^2 + d_y^2 = |f|^2$  is unchanged.

As a sum of sines,  $d_z$  is odd,  $d_z(-\mathbf{q}) = -d_z(\mathbf{q})$ , consistent with broken time reversal by Eq. (S26). At  $\mathbf{K} = (-\frac{2\pi}{3\sqrt{3}}, \frac{2\pi}{3})$  the three phases are

$$\mathbf{K} \cdot \mathbf{n}_1 = \frac{2\pi}{3}, \quad \mathbf{K} \cdot \mathbf{n}_2 = \frac{4\pi}{3}, \quad \mathbf{K} \cdot (\mathbf{n}_1 - \mathbf{n}_2) = -\frac{2\pi}{3}, \quad (\text{S43})$$

so

$$d_z(\mathbf{K}) = 6\kappa \left[ \sin \frac{2\pi}{3} + \sin \frac{4\pi}{3} + \sin \left( -\frac{2\pi}{3} \right) \right] = 6\kappa \left( \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} \right) = -3\sqrt{3}\kappa, \\ \Delta = 2 |d_z(\mathbf{K})| = 6\sqrt{3} |\kappa|. \quad (\text{S44})$$

By oddness  $d_z(-\mathbf{K}) = +3\sqrt{3}\kappa$ , so the two Dirac points have opposite masses,

$$m_{\pm\mathbf{K}} \equiv d_z(\pm\mathbf{K}) = \mp 3\sqrt{3}\kappa. \quad (\text{S45})$$

This is the same mechanism as the Haldane model [2]. The gap opening is shown in Fig. S5.

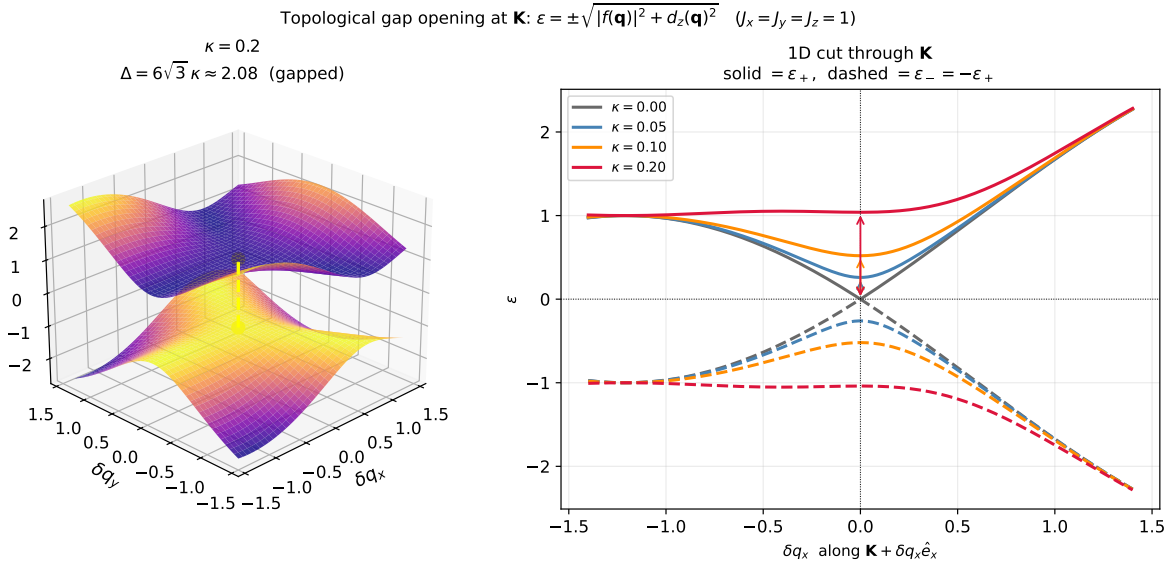


FIG. S5. Topological gap opening at the Dirac point  $\mathbf{K}$ . *Left*: the field-induced mass  $d_z(\mathbf{q})$  gaps the dispersion  $\varepsilon_{\pm} = \pm \sqrt{|f|^2 + d_z^2}$ . *Right*: a one-dimensional cut through  $\mathbf{K}$  for increasing  $\kappa$ , with  $\varepsilon_+$  (solid) and  $\varepsilon_-$  (dashed) splitting apart as the gap opens.

## SVIII. CALCULATION OF CHERN NUMBER

The invariant of the gapped phase is the Chern number of the occupied band [3],

$$\mathcal{C} = \frac{1}{2\pi} \int_{\text{BZ}} \Omega(\mathbf{q}) d^2q, \quad \Omega = \partial_{q_x} A_y - \partial_{q_y} A_x, \quad A_\mu(\mathbf{q}) = -i \langle u_- | \partial_{q_\mu} u_- \rangle, \quad (\text{S46})$$

with  $|u_- \rangle$  the normalized negative-energy eigenstate of  $h(\mathbf{q})$ .

Parametrize the unit vector as

$$\hat{\mathbf{d}} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta), \quad \cos \theta = \frac{d_z}{|\mathbf{d}|}, \quad e^{i\phi} = \frac{d_x + i d_y}{\sqrt{d_x^2 + d_y^2}}, \quad (\text{S47})$$

where  $d_x + i d_y = -i f^*(\mathbf{q})$  by Eq. (S11). The lower-band eigenstate in one gauge is

$$|u_- \rangle = \begin{pmatrix} \sin \frac{\theta}{2} e^{-i\phi} \\ -\cos \frac{\theta}{2} \end{pmatrix}, \quad (\text{S48})$$

which satisfies  $(\hat{\mathbf{d}} \cdot \boldsymbol{\sigma}) |u_- \rangle = -|u_- \rangle$ . One can check this by direct multiplication using  $\hat{\mathbf{d}} \cdot \boldsymbol{\sigma} = \begin{bmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{bmatrix}$ . Here I choose this gauge because it is smooth at the north pole  $\theta = 0$ ; it is singular at  $\theta = \pi$ , but a second patch covers that point, and  $\mathcal{C}$  does not depend on the patch. With  $s = \sin \frac{\theta}{2}$ ,  $c = \cos \frac{\theta}{2}$ ,

$$\partial_\mu |u_- \rangle = \begin{pmatrix} e^{-i\phi} (\frac{1}{2} c \partial_\mu \theta - i s \partial_\mu \phi) \\ \frac{1}{2} s \partial_\mu \theta \end{pmatrix}, \quad \langle u_- | = (s e^{+i\phi}, -c). \quad (\text{S49})$$

Term by term,

$$\begin{aligned} \langle u_- | \partial_\mu u_- \rangle &= s e^{i\phi} e^{-i\phi} (\frac{1}{2} c \partial_\mu \theta - i s \partial_\mu \phi) - c (\frac{1}{2} s \partial_\mu \theta) \\ &= \frac{1}{2} s c \partial_\mu \theta - i s^2 \partial_\mu \phi - \frac{1}{2} s c \partial_\mu \theta = -i s^2 \partial_\mu \phi. \end{aligned} \quad (\text{S50})$$

The  $\frac{1}{2} s c \partial_\mu \theta$  terms cancel, so

$$A_\mu = -i \langle u_- | \partial_\mu u_- \rangle = -s^2 \partial_\mu \phi = -\frac{1}{2} (1 - \cos \theta) \partial_\mu \phi. \quad (\text{S51})$$

Take the curl with  $g \equiv \frac{1}{2} (1 - \cos \theta)$  and  $\partial_\mu g = \frac{1}{2} \sin \theta \partial_\mu \theta$ . The mixed second derivatives of  $\phi$  cancel, so

$$\Omega = -[(\partial_x g)(\partial_y \phi) - (\partial_y g)(\partial_x \phi)] = -\frac{1}{2} \sin \theta (\partial_x \theta \partial_y \phi - \partial_y \theta \partial_x \phi) = -\frac{1}{2} \hat{\mathbf{d}} \cdot (\partial_{q_x} \hat{\mathbf{d}} \times \partial_{q_y} \hat{\mathbf{d}}). \quad (\text{S52})$$

The last equality is the solid-angle Jacobian identity  $\hat{\mathbf{d}} \cdot (\partial_x \hat{\mathbf{d}} \times \partial_y \hat{\mathbf{d}}) = \sin \theta (\partial_x \theta \partial_y \phi - \partial_y \theta \partial_x \phi)$ .

The Chern number is (minus) the degree of the map  $\mathbf{q} \mapsto \hat{\mathbf{d}}$ ,

$$\mathcal{C} = -\frac{1}{4\pi} \int_{\text{BZ}} \hat{\mathbf{d}} \cdot (\partial_{q_x} \hat{\mathbf{d}} \times \partial_{q_y} \hat{\mathbf{d}}) d^2 q. \quad (\text{S53})$$

For the gapped cones we also need the in-plane winding

$$\nu_i = \frac{1}{2\pi} \oint_{\mathcal{C}_i} d(\arg f(\mathbf{q})) \quad (\text{S54})$$

around each Dirac point. From the linearization Eq. (S17),  $f(\mathbf{K} + \delta \mathbf{q}) \propto (\delta q_x - i \delta q_y)$ . Its phase winds by  $-2\pi$  on a counterclockwise loop, so  $\nu_{\mathbf{K}} = -1$ . At  $-\mathbf{K}$ ,  $f(-\mathbf{K} + \delta \mathbf{q}) = f(\mathbf{K} - \delta \mathbf{q})^* \propto (\delta q_x + i \delta q_y)$ , so  $\nu_{-\mathbf{K}} = +1$ . The two windings are opposite: the total winding over the Brillouin-zone torus must vanish (Nielsen–Ninomiya [4]), so Dirac points come in pairs of opposite chirality.

Near  $\mathbf{K}_i$  write  $f \simeq v e^{i\beta_i} (\delta q_x + i \nu_i \delta q_y)$  with  $v = \frac{3J}{2}$ . The constant phase  $\beta_i$  only rotates  $(d_x, d_y)$  in the plane and drops out of the triple product, so we set  $\beta_i = 0$ . With  $m_i = d_z(\mathbf{K}_i)$  constant to leading order,

$$\mathbf{d}(\mathbf{K}_i + \delta \mathbf{q}) = (-\text{Im } f, -\text{Re } f, m_i) = (-v \nu_i \delta q_y, -v \delta q_x, m_i). \quad (\text{S55})$$

The triple product is constant:

$$\begin{aligned} \partial_x \mathbf{d} &= (0, -v, 0), & \partial_y \mathbf{d} &= (-v \nu_i, 0, 0) \implies \partial_x \mathbf{d} \times \partial_y \mathbf{d} = (0, 0, -v^2 \nu_i), \\ \mathbf{d} \cdot (\partial_x \mathbf{d} \times \partial_y \mathbf{d}) &= -v^2 \nu_i m_i. \end{aligned} \quad (\text{S56})$$

Normalizing by  $|\mathbf{d}|^3 = (v^2 \rho^2 + m_i^2)^{3/2}$  and integrating in polar coordinates  $\delta \mathbf{q} = \rho(\cos \alpha, \sin \alpha)$ ,

$$\begin{aligned} \mathcal{C}_i &= -\frac{1}{4\pi} \int \frac{-v^2 \nu_i m_i}{(v^2 \rho^2 + m_i^2)^{3/2}} d^2 \delta q = \frac{\nu_i m_i}{4\pi} (2\pi) \int_0^\infty \frac{v^2 \rho d\rho}{(v^2 \rho^2 + m_i^2)^{3/2}} \\ &= \frac{\nu_i m_i}{2} \left[ \frac{-1}{\sqrt{v^2 \rho^2 + m_i^2}} \right]_0^\infty = \frac{1}{2} \nu_i \text{sgn}(m_i). \end{aligned} \quad (\text{S57})$$

Each cone contributes half a unit of Chern number. Summing,

$$\mathcal{C} = \frac{1}{2} \sum_{i=\pm \mathbf{K}} \text{sgn}(m_i) \nu_i. \quad (\text{S58})$$

From Eq. (S45),  $\text{sgn}(m_{\pm \mathbf{K}}) = \mp \text{sgn}(\kappa)$ , and  $\nu_{\pm \mathbf{K}} = \mp 1$  from above:

$$\mathcal{C} = \frac{1}{2} \left[ (-\text{sgn } \kappa)(-1) + (+\text{sgn } \kappa)(+1) \right] = \text{sgn}(\kappa) = \text{sgn}(h^x h^y h^z), \quad \boxed{\mathcal{C} = \pm 1 = \text{sgn}(\kappa).} \quad (\text{S59})$$

If  $d_z$  is even (the time-reversal-symmetric case of Eq. (S26)), the two masses are equal,  $m_{-\mathbf{K}} = m_{\mathbf{K}}$ . The windings still obey  $\nu_{\mathbf{K}} = -\nu_{-\mathbf{K}}$ . Then

$$\mathcal{C} = \frac{1}{2} \text{sgn}(m_{\mathbf{K}})(\nu_{\mathbf{K}} + \nu_{-\mathbf{K}}) = 0. \quad (\text{S60})$$

The two half-integer contributions cancel and the gap is trivial. This confirms Section SIV: a nonzero Chern number requires the time-reversal breaking mass.

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